

# Autonomous On-Board Human Activity Recognition, Orientation-Agnostic 3D Reconstruction and Deterministic Sequence Validation

for Microgravity Payload Experiments

Final Project Documentation: Main Idea and Approach

|                    |                                                                           |
|--------------------|---------------------------------------------------------------------------|
| Problem Title      | AI Human Activity Recognition for On-board BAS Experiments                |
| Organisation       | Indian Space Research Organisation (ISRO), Department of Space            |
| Category / Theme   | Software   Space Technology                                               |
| Target Environment | Bharatiya Antariksh Station: BAS-03 / BAS-04 science modules              |
| Solution Type      | Offline, edge-first, multi-agent AI system with a deterministic validator |
| Project Status     | Design-complete; partially implemented prototype                          |

# 1. Executive Summary

## 1.1 The problem

Scientific experiments aboard an orbital laboratory are procedure-critical. A step performed out of order, or skipped, can invalidate an experiment that cost years of ground preparation and a launch slot to place in orbit. On Earth, compliance is assured by a Principal Investigator watching a live video feed. In orbit this model breaks down for three structural reasons: downlink bandwidth is scarce and cannot carry continuous high-definition video; ground-station visibility is intermittent; and for lunar and deeper missions, light-time delay makes real-time supervision impossible.

ISRO Problem Statement 26174 therefore asks for an **on-board, offline, standalone AI system** that watches a fixed payload camera, tracks the sequence of a predefined experiment, tells the astronaut what to do next, raises a **voice alert** when a step is skipped or performed out of order, writes a **timestamped structured log**, records video locally, streams it to a specified IP, and shows everything in a monitoring GUI, with no cloud dependency at all.

The statement also poses an optional research challenge. Terrestrial pose models assume a gravity-aligned "up", which astronauts do not have. The system should therefore track the body **relative to the payload rack, not the floor**, using orientation-agnostic 3D Human Mesh Recovery (HMR).

## 1.2 Why the problem is hard

| Difficulty         | What goes wrong                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Orientation</b> | Pose networks carry an implicit upright-body prior. An inverted or floating subject causes left/right limb confusion and tracking collapse under self-occlusion.                         |
| <b>Posture</b>     | A relaxed body in microgravity adopts the Neutral Body Posture, whose joint angles differ systematically from terrestrial training data.                                                 |
| <b>Geometry</b>    | Deciding whether a hand grasped a box or merely passed near it is a metric 3D question. A single camera does not measure depth, so it must be reconstructed with quantified uncertainty. |
| <b>Trust</b>       | A validator that can hallucinate is worse than none, because it gives confident wrong verdicts. Generative components must be prevented from issuing the verdict.                        |

## 1.3 The proposed solution

An **offline, edge-first, multi-agent system** built around a shared Digital Twin and terminated by a deterministic validator. Eight specialised agents (Perception and HMR, IMU, Fusion, HAR, Digital Twin, Validation, Reasoning, Monitoring) run asynchronously against a thread-safe shared blackboard instead of a rigid call chain. A stall in one agent therefore degrades that agent's contribution rather than halting the pipeline.

**Camera** → video acquisition → **perception** (YOLO object detection, orientation-agnostic 3D HMR, AdaSpot RoI gating) → **2D-to-3D reconstruction** (camera frame → payload-rack frame) → **hand-object geometry** (distance, contact, grasp, extraction) → **Digital Twin** → **deterministic FSM and graph validator** → voice alert, GUI, telemetry, local recording and network stream

### The central architectural commitment

**Neural components perceive. A deterministic automaton decides.** The local VLM/LLM and the Reasoning Agent may interpret, explain and recommend, but they may never declare a step complete, declare an experiment successful, or suppress an anomaly. That authority belongs only to a finite-state machine whose every decision can be reconstructed from logged evidence.

## 1.4 What is new

The individual algorithms (YOLO, HMR, Kalman filtering, FSMs, TTS) are established. The contribution is at the systems level:

1. **Orientation-agnosticism as a coordinate-frame problem.** All kinematics are transformed into a fixed payload-rack frame, rather than relying on data augmentation.
2. **Metric 3D geometry as evidence.** A grasp is defined by a distance threshold with stated uncertainty, not by a classifier confidence score.
3. **The Digital Twin as the integration contract** between eight decoupled agents, so each is independently testable and the run is replayable from telemetry.
4. **An enforced safety boundary** between generative reasoning and authoritative validation.
5. **Structured event telemetry as the primary downlink product**, with video kept locally, which matches the real bandwidth economics of orbit.

## 1.5 Current status

A working prototype exists: a Python orchestrator, an eight-agent pipeline over a shared-memory blackboard, a browser-based mission-control dashboard with live MJPEG streaming, an FSM validating the red/yellow-box experiment, offline TTS, and JSONL telemetry logging. The orientation-agnostic HMR path, the IMU and UKF fusion path, the local VLM/LLM layer and the graph-guided validator are designed in detail but **not yet complete**. No component has been flight-qualified or tested in real microgravity, and this document makes no such claim.

# 2. Problem Statement and Requirements

## 2.1 The statement as issued

| Field                | Value                                                      |
|----------------------|------------------------------------------------------------|
| Problem Statement ID | 26174                                                      |
| Title                | AI Human Activity Recognition for On-board BAS Experiments |
| Organisation         | Indian Space Research Organisation (ISRO)                  |
| Department           | Department of Space / ISRO                                 |
| Category             | Software                                                   |
| Theme                | Space Technology                                           |

## 2.2 Requirements

The expected solution decomposes into eight functional requirements, one optional research requirement and one deliverable. Each carries an identifier that is used throughout this document and traced to a component later.

| ID        | Requirement (as stated)                                                                                   | Engineering interpretation                                                                                              |
|-----------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>R1</b> | Continuously process local video feeds to track the sequence of the experiment                            | Sustained, unattended, real-time inference on a fixed-camera stream with no operator intervention                       |
| <b>R2</b> | At the start, or after each step, suggest the next step                                                   | Guidance derived from the validated procedure state, not from a static script                                           |
| <b>R3</b> | Alert when a step is skipped or an out-of-sequence step is added; the alert must be voice-based           | Detect two distinct anomaly classes, synthesise speech locally, and alert fast enough to interrupt the erroneous action |
| <b>R4</b> | Generate a timestamped, structured, lightweight text file of steps with outcomes                          | Append-only, machine-parseable, auditable, compact event log                                                            |
| <b>R5</b> | Stream the video to a specific IP                                                                         | Network video service on the local segment                                                                              |
| <b>R6</b> | Store the video locally                                                                                   | Independent local recording path, not derived from the stream                                                           |
| <b>R7</b> | Graphical user interface for monitoring                                                                   | Unified operator console for all of the above                                                                           |
| <b>R8</b> | Dataset generation for object detection, pose estimation and hand-object interaction                      | Custom, task-focused corpus; synthetic generation is expected                                                           |
| <b>R9</b> | (Optional) Orientation-agnostic 3D HMR tracking the astronaut relative to the payload rack, not the floor | The research core of the problem                                                                                        |
| <b>D1</b> | (Deliverable) A trained AI model that runs on an offline standalone system                                | Zero cloud dependency at inference time                                                                                 |

## 2.3 The benchmark experiment

The problem statement supplies a sample procedure that is used as the reference scenario throughout this document. A container box holds two smaller boxes, one **red** and one **yellow**. The procedure is:

1. Open the container.
2. Extract the red box.
3. Extract the yellow box.

The scenario is deliberately simple, and that is useful for three reasons. It is short enough to demonstrate live in front of a jury. It contains an unambiguous ordering constraint (red before yellow), so out-of-order detection can be shown on demand. And it exercises every layer of the stack: detection, 3D reconstruction, hand-object geometry, temporal persistence, state transition, guidance, alerting and logging.

**Scope of this document**

The remaining chapters explain the main idea and the approach: how each requirement above is met, the architecture behind it, and how the result will be evaluated. Features that are designed but not yet built are marked as planned.

### 3. Mission Context and Problem Analysis

#### 3.1 Mission context: the Bharatiya Antariksh Station

Under the Gaganyaan programme and the Amrit Kaal Vision 2047 roadmap, ISRO is developing the Bharatiya Antariksh Station (BAS), a modular orbital laboratory. Publicly stated targets are a first base module (BAS-01) in December 2028 on an LVM3, and a five-module, permanently crewed station of about 52 tonnes by 2035, at 400–450 km altitude and 51.5° inclination. These figures come from public ISRO communications and secondary sources, not from us.

| Module         | Role and relevance                                                                                                             |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| BAS-03         | <b>Science research module:</b> microgravity biotechnology, physical sciences, human physiology. <b>Target of this system.</b> |
| BAS-04         | <b>Laboratory module:</b> extended experimental racks and technology demonstrations. <b>Target of this system.</b>             |
| BAS-01, 02, 05 | Base, core and common berthing modules (crew quarters, station control, logistics). Not procedure-critical science.            |

The design assumes a **fixed, rack-mounted monocular RGB camera**; one astronaut in the workspace at a time; artificial cabin lighting with hard shadows; cluttered rack surfaces and cabling; an astronaut in **arbitrary orientation, including inverted**; a local station LAN with no internet; and continuous multi-hour operation.

#### 3.2 The communications constraint and its consequences

This system exists because of a bandwidth and availability problem, not because autonomy is fashionable. Three independent constraints apply:

- **Bandwidth.** Continuous HD video competes with every other station system for S-band and Ka-band downlink capacity.
- **Routing delay.** Relay through tracking and data relay satellites degrades instant ground feedback, even in low Earth orbit.
- **Loss of signal.** Ground visibility is periodic, and during these windows there is no real-time external support.

For lunar and cis-lunar missions the limit becomes absolute: light-time delay makes real-time supervision impossible however much bandwidth exists. If a Principal Investigator cannot watch, the procedure must be validated on board. That yields three criteria that drive the whole architecture: **fully offline inference** (all weights resident locally); **hands-free interaction** (an astronaut with both hands in a glovebox cannot read a screen, so guidance must be audible and generated locally); and **bandwidth-proportionate reporting** (kilobytes of structured telemetry downlinked, video kept on board).

#### 3.3 Problem analysis: why the naive approach fails

Each row below names a failure of the obvious approach and the requirement it creates. Later chapters answer each requirement.

| Challenge                   | Why the naive approach fails                                                                                                                                                                                                                                                                 | Requirement generated                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Gravity prior</b>        | Models trained on upright subjects absorb the assumption that the head is above the hips. For inverted subjects they confuse left and right limbs, read global rotation as joint deformation, and collapse under occlusion. A confidently mislocated hand gives a confidently wrong verdict. | Orientation must be <b>factored out</b> of the representation, not merely tolerated.                                   |
| <b>Neutral Body Posture</b> | A relaxed body in microgravity settles into a different postural regime. Reported deviations from standing include about 39° shoulder flexion, 77° elbow flexion and 55° hip and knee flexion. Models shaped by terrestrial data run outside their training distribution.                    | Training data must contain <b>microgravity-plausible postures</b> (captured, synthesised or adapted).                  |
| <b>Monocular depth</b>      | One camera maps a 3D ray to a 2D pixel, so depth is not observable. Learned body-scale priors give an <b>estimate</b> , whose error grows with distance, occlusion and unusual poses.                                                                                                        | Report distances as <b>estimates with characterised error</b> , never as exact, and set thresholds against that error. |
| <b>Verification</b>         | A network output such as "extract red box, 0.94" is a softmax score. It cannot be audited, cannot be decomposed into evidence, and cannot be certified.                                                                                                                                      | The verdict must come from a <b>deterministic, replayable mechanism</b> over recorded evidence.                        |
| <b>Hallucination</b>        | A VLM will answer "has the red box been extracted?" fluently, whether or not it has enough evidence. Fluent wrong answers are trusted, so they are dangerous.                                                                                                                                | The generative layer must be <b>architecturally prevented</b> from issuing verdicts.                                   |
| <b>Compute</b>              | Running full-resolution HMR, detection and temporal models on every frame saturates the edge thermal budget. In orbit there is no convection, so sustained throttling is realistic and silently degrades validation.                                                                         | Computation must be <b>gated by relevance</b> , not applied uniformly.                                                 |

### Takeaway

The hard part is not recognising a person. It is producing a **trustworthy, auditable procedural verdict** from an uncertain, single-camera view of a body in an unfamiliar posture, on limited on-board hardware, with no ground support.

## 4. Existing Approaches and Our Positioning

### 4.1 Survey of conventional approaches

Six families of methods could be applied to this problem. Each has real strengths, but none alone can produce a metric, auditable, orientation-independent verdict on a procedure.

| Approach                                                                                      | What it does well                                                                   | Why it is not sufficient alone                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2D HAR from RGB video</b> (3D CNNs, TSN/TSM, video transformers)                           | Mature and accurate on terrestrial benchmarks; emits an activity label from a clip. | Produces no geometry. It cannot say how far a hand was from a box, cannot separate reaching-past from grasping, inherits the upright prior, and its label cannot be decomposed into evidence. |
| <b>2D pose estimation</b> (OpenPose, HRNet, ViTPose, MediaPipe)                               | Fast, accurate pixel-space keypoints with good tooling.                             | Pixel distance is not metric distance: a hand 30 cm behind a box can project right next to it. Contact cannot be judged without depth, and the upright prior remains.                         |
| <b>Conventional 3D pose</b> (lifting or direct 3D regression)                                 | Gives 3D joints of the body.                                                        | Usually root-relative and scale-normalised. Hand-to-object distance needs absolute placement in a common frame, which these outputs do not provide.                                           |
| <b>Instrumented objects</b> (RFID, IMU-tagged boxes, smart containers)                        | Highly reliable sequence detection.                                                 | Requires modifying flight hardware and adds mass, power and certification burden for every experiment. It does not generalise to arbitrary payloads.                                          |
| <b>LLM/VLM as validator</b>                                                                   | Appears to solve everything with one component.                                     | It answers fluently without calibrated confidence, so it can issue confident wrong verdicts (Chapter 3, hallucination).                                                                       |
| <b>Pure rule-based computer vision</b> (colour thresholds, background subtraction, tripwires) | Deterministic and auditable.                                                        | Brittle under cabin lighting, clutter and occlusion. It is trustworthy but not capable.                                                                                                       |

The last two rows fail in opposite ways: the VLM is capable but not trustworthy, while rule-based vision is trustworthy but not capable. Our architecture takes **determinism from the rule-based family** and **perceptual capability from the 2D/3D vision family**, and keeps them in separate layers.

### 4.2 Our positioning

We do not claim to have invented orientation-agnostic HMR, saliency-gated inference, visual-inertial fusion or finite-state validation. Each has its own literature. The claim is narrower and defensible:

#### Our contribution

The **integration of metric 3D human-object geometry with deterministic procedural validation**, mediated by a **Digital Twin**, inside a **fully offline edge system**, addressed specifically to **microgravity payload operations**.

## 5. Proposed Solution and Design Principles

### 5.1 Solution overview

A fixed monocular payload camera feeds an offline edge computer. Object detection localises the experiment items, and orientation-agnostic Human Mesh Recovery reconstructs the astronaut's 3D body and hand joints. Both are transformed into a fixed **payload-rack coordinate frame R**, so geometry is expressed relative to the workbench, not to the floating body or the camera. Metric hand-object distances in R, accumulated over time, yield interaction events: approach, contact, grasp, extract and release. These events are written into a **Digital Twin**, the single source of truth for eight decoupled agents. A **deterministic FSM**, later extended by a graph validator, compares the twin against the authorised procedure and issues the verdict: valid, out-of-order or skipped. A Reasoning Agent and a local VLM/LLM explain and guide from the validated state but cannot alter it.

### 5.2 Solution pillars

| Pillar                                                                        | Addresses   |
|-------------------------------------------------------------------------------|-------------|
| Edge-first offline inference                                                  | D1          |
| Orientation-agnostic 3D HMR in rack frame R                                   | R9          |
| Metric hand-object geometry as evidence                                       | R1          |
| Digital Twin and eight-agent orchestration                                    | Integration |
| Deterministic FSM as sole authority; graph validator for branching procedures | R1, R3      |
| Local VLM/LLM, advisory only                                                  | R2          |
| Offline TTS voice guidance and alerts                                         | R2, R3      |
| JSONL telemetry as primary downlink product                                   | R4          |
| Dual video path: local record and LAN stream                                  | R5, R6      |
| Mission-control GUI                                                           | R7          |
| Synthetic and real dataset engine                                             | R8          |

### 5.3 What the astronaut and mission control experience

No dashboard interaction is needed to perform the procedure. The astronaut hears:

*"System initialised. Please open the primary container box."*

*"Container opened. Next step: please extract the red box."*

*"Warning: procedural error. The red box must be extracted before the yellow box."*

*"Yellow box extracted. Experiment complete."*

The screen is confirmatory only. Mission control receives **not video** but a JSONL file appended in real time, where each line is a timestamped event with state, evidence, confidence and outcome. The whole session can be audited, and replayed against the same validator to reproduce every verdict. Video stays on board.

## 5.4 Headline objectives and explicit boundaries

Of the seventeen functional objectives, these carry the strongest acceptance criteria:

- **Reproducibility:** an identical evidence sequence must give an identical verdict sequence (100%).
- **Latency:** anomaly-to-voice-alert time of 1.5 s or less, including debounce.
- **Offline:** the system must run with the network interface physically disabled.
- **Frame consistency:** a static object's rack-frame coordinates stay constant as the camera moves, within calibration error.
- **Distance accuracy:** hand-object distance error is measured (MAE, RMSE) against ground truth.

The system is **not** a general activity recogniser, **not** a safety-of-life system, **not** an autonomous decision-maker (the crew can override or ignore it), **not** a video downlink, and **not** flight-qualified or microgravity-validated.

## 5.5 Design principles

Seven principles govern every later design decision. Each later chapter names the principle that forced its choice.

| Principle                                        | Meaning                                                                           | Consequence                                                                                  |
|--------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>DP-1 Edge-first</b>                           | All inference is local; no runtime network need.                                  | Model size and quantisation are constrained; a local VLM replaces any API.                   |
| <b>DP-2 Orientation-agnostic by construction</b> | Achieved by changing the coordinate frame, not by hoping the network generalises. | Camera-to-rack calibration is mandatory; its error becomes a first-class error term.         |
| <b>DP-3 Deterministic authority</b>              | The verdict-issuing component is finite, enumerable and replayable.               | The FSM consumes thresholded, debounced, logged evidence, never raw neural confidence alone. |
| <b>DP-4 Evidence before assertion</b>            | No state changes on a single frame.                                               | Debouncing is architectural, not a tuning afterthought.                                      |
| <b>DP-5 Decoupled agents over a shared twin</b>  | Agents communicate only through the Digital Twin.                                 | Any agent can stall or be replaced; the system degrades partially, not totally.              |
| <b>DP-6 Auditability</b>                         | Every verdict is reconstructible from telemetry alone.                            | The triggering evidence is logged with each transition.                                      |
| <b>DP-7 Honest uncertainty</b>                   | Estimates are labelled as estimates and carry error.                              | Thresholds cite measured error; low-confidence states are declared, not guessed.             |

## 6. System Architecture and Multi-Agent Design

### 6.1 Layered architecture

The system has six layers. Data flows downward from sensing to output; authority is granted only at the decision layer. Figure 6.1 shows the components in each layer.

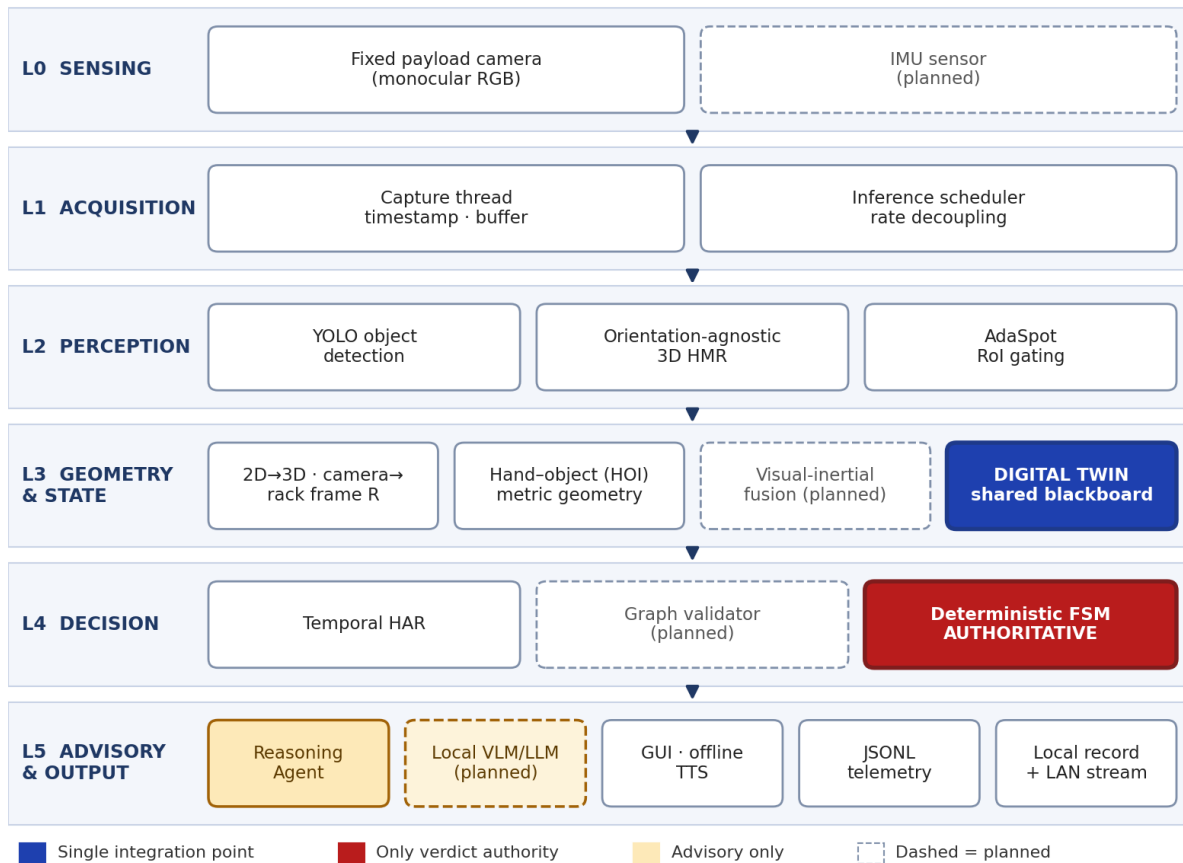


Figure 6.1: Layered system architecture (dashed boxes are planned, not yet built)

Two elements are deliberately highlighted. The **Digital Twin** (blue) is the only integration point, so no agent calls another agent directly. The **FSM** (red) is the only component allowed to issue a verdict. Advisory components (amber) read validated state and write text; they have no write path to the verdict.

### 6.2 Authority model

**Neural perception** (probabilistic) → **thresholded evidence** (with confidence and timestamp) → **Digital Twin** (structured state) → **deterministic validation** (FSM, debounced) → **authorised transition** (logged with evidence) → **guidance, alert and telemetry**

The VLM, LLM and Reasoning Agent may explain the outcome, but they **may not write to the validator**. This is enforced in code: the FSM's transition function accepts only Digital Twin evidence fields and holds no reference to any language-model output. The chapter on the deterministic validation core covers this in depth.

### 6.3 Multi-agent orchestration

A single sequential pipeline would be unacceptable here for three reasons. **Rate coupling:** a 128 Hz IMU and a 12 Hz HMR cannot share one loop without one being wrong. **Fault coupling:** a failing TTS engine would halt validation, which is absurd because validation is the important part. **Test coupling:** the safety-critical component would only be testable in integration. Independent agents on a shared blackboard remove all three. Eight agents make up the system:

| Agent                        | Role                                                                                                | Safety impact | Status                           |
|------------------------------|-----------------------------------------------------------------------------------------------------|---------------|----------------------------------|
| <b>1. Perception and HMR</b> | Detect objects, reconstruct the astronaut in 3D, transform everything to rack frame R               | High          | Detection built; HMR in progress |
| <b>2. IMU</b>                | Acquire, timestamp and bias-correct inertial data (128 Hz)                                          | Low           | Planned                          |
| <b>3. Fusion</b>             | Visual-inertial UKF estimate with uncertainty                                                       | Medium        | Planned                          |
| <b>4. HAR</b>                | Derive hand-object events and activity labels                                                       | Medium-high   | Prototype                        |
| <b>5. Digital Twin</b>       | Keep the twin consistent, serve atomic snapshots, decay stale data                                  | High          | Blackboard built                 |
| <b>6. Validation</b>         | Evaluate twin against the procedure and issue the verdict (exclusive writer of the procedure layer) | Critical      | FSM built                        |
| <b>7. Reasoning</b>          | Turn validated state into guidance and explanation; advisory only                                   | Low           | Templates built; VLM planned     |
| <b>8. Monitoring</b>         | GUI, voice, telemetry, local recording and LAN streaming                                            | Low           | Built                            |

Earlier team papers described four agents in one place and eight in another. The four-agent view is only a briefing simplification; the authoritative architecture has **eight** agents.

### 6.4 The Digital Twin

The Digital Twin is a **structured state object**, continuously synchronised with perception, that any agent can read and designated agents can write. It is not just a 3D animation: the rendering is a view of the twin, while the twin itself is the data. A structured model can be validated against, logged, replayed and tested, and that is what this system needs.

| Layer              | Contents                                                            | Written by                     |
|--------------------|---------------------------------------------------------------------|--------------------------------|
| <b>Static</b>      | Rack geometry, workspace volume, camera pose, calibration transform | Calibration (once per session) |
| <b>Kinematic</b>   | 3D joints and hand joints in R, body mesh, velocities, covariances  | Perception, Fusion             |
| <b>Object</b>      | Per-object id, class, position in R, state, track age, colour check | Perception                     |
| <b>Interaction</b> | Distance matrices, hand-object state per object, activity label     | HAR                            |
| <b>Procedure</b>   | FSM state, debounce counters, anomaly state, step history           | <b>Validation only</b>         |

| Layer  | Contents                                                             | Written by              |
|--------|----------------------------------------------------------------------|-------------------------|
| Health | Inference rate, dropped frames, heartbeats, CPU/GPU/thermal, storage | All agents (own fields) |

- **Single-writer rule.** Only the Validation Agent may write the procedure layer, so every verdict traces to exactly one component.
- **Timestamps and staleness.** Every field carries its own timestamp and confidence. Data older than about two inference periods is treated as stale and no longer usable for transitions; beyond five periods it is expired. Confidence decays automatically, so careless use is also safe use.
- **Testability.** Because the FSM consumes only the twin, the whole validation layer can be tested with synthetic twin sequences and no camera, model or human. The safety-critical part becomes the most testable part.
- **Replay and sim-to-real.** A session can be re-run offline from telemetry, and synthetic data and live perception fill the same twin schema, so validators run unchanged on real data.

## 6.5 Deployment architecture

The **current prototype** runs on a development PC (Python, CPU) with a USB webcam or recorded video, a browser dashboard on localhost, and local .jsonl and .mp4 outputs. The **planned edge target** is an NVIDIA Jetson Orin NX for vision, HMR, TTS and VLM, paired with a separate deterministic co-processor for the FSM, telemetry and watchdog, plus on-board NVMe recording and a station-LAN stream. The split is architectural: perception and validation sit in **separate fault domains**, so a crash or driver hang in the vision stack cannot corrupt state tracking or logging. In the prototype this separation is logical (threads and processes); on the target it would become physical.

## 7. Video Acquisition and Object Perception

### 7.1 Video acquisition pipeline

The pipeline turns a continuous camera signal into timestamped, rate-controlled frames for inference, while a separate full-rate path serves recording and streaming. Input is the fixed monocular RGB camera at 30 FPS, or a recorded file for development and regression tests. The main design decisions are:

| Decision                                        | Reason                                                                                                                                                                                                                                       |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dedicated capture thread</b>                 | Capture never blocks on downstream consumers. Each frame is stamped immediately with a <b>monotonic clock</b> (wall-clock time can step backwards under NTP correction and corrupt interval arithmetic) and carries a never-reused frame id. |
| <b>Bounded ring buffer, drop-oldest</b>         | An unbounded queue is a memory-exhaustion bug on a multi-hour session. Drop-oldest is used because for live guidance a stale frame has negative value.                                                                                       |
| <b>Three separate paths</b>                     | Recording (must not drop), LAN streaming (may drop) and inference (decimated to 10–15 FPS by the scheduler) all read views of the same buffer.                                                                                               |
| <b>Letterbox preprocessing with inverse map</b> | Aspect-preserving resize avoids distorting limb geometry. The scale and padding are kept with each frame so every detection maps back to original pixels; skipping this causes a systematic bias that is fatal to metric distance.           |
| <b>Alignment by timestamp, never by index</b>   | Index alignment silently breaks the first time a frame is dropped. Video, IMU samples and events are synchronised by time.                                                                                                                   |
| <b>Gaps are never silently absorbed</b>         | Every dropped frame is counted in health telemetry. A gap longer than the debounce window resets the persistence counters, so evidence from before and after a gap is not treated as continuous.                                             |

**Operating point.** Higher inference rate, resolution and model size all cost power and heat, and thermal throttling then reduces the effective frame rate. We therefore record at full rate but infer at 10–15 FPS, gate high-resolution work by saliency, and specify debounce in seconds. This is justified because the events being detected are human manipulation actions with sub-hertz onset dynamics: 30 FPS inference would buy temporal resolution the task does not need at the cost of resolution it does need. If the camera fails, the twin state is frozen, FSM transitions are suspended and an alert is raised.

### 7.2 Object perception

The goal is not bounding boxes but **structured, tracked, temporally consistent object observations**. The detector is **YOLOv8-nano**, chosen for its accuracy per watt on edge hardware and its mature INT8/TensorRT export path (running at FP32 in the prototype; quantised edge deployment is planned). It detects six classes: container, container lid (tracked separately because lid displacement triggers an FSM transition), red box, yellow box, hand, and a rack fiducial marker for calibration. Rack structure and the astronaut's body are handled by the HMR path and the static rack model.

#### From detector output to observation

decode and NMS → undo letterbox → class-conditional confidence gate → **colour verification** → track association (IoU, motion, identity) → temporal smoothing → 2D-to-3D lift into rack frame R → object state estimate (docked, in container, touched, grasped, extracted, lost)

Each observation record holds a stable object id, class, pixel box, position in R with its estimated uncertainty (sigma), calibrated confidence, a colour-verified flag, state, track age and an occlusion flag.

**The FSM consumes state, position and track age, never a raw class logit.**

- **Colour-aware identification.** The red and yellow boxes are geometrically identical, and cabin lighting produces coloured shadows and highlights. A confident red/yellow swap is the worst possible failure, because the system would report the correct sequence while the astronaut performs the wrong one. A separate HSV check on the eroded central region of each box therefore confirms the learned class. On disagreement, confidence is downgraded and the FSM transition is **blocked**, reported as an ambiguous-object event with a spoken request to reposition. Asking is cheap; a wrong verdict is not.
- **Confidence calibration.** Raw detector scores are not probabilities. They are calibrated on a validation split (temperature scaling or isotonic regression, under evaluation) so that the evidence threshold has a stated meaning.
- **Tracking.** A lightweight tracker links detections by IoU and centroid motion, with class consistency as a hard rule: a red-box track can never be matched to a yellow-box detection.

| Failure                            | Response                             | Safety consequence                  |
|------------------------------------|--------------------------------------|-------------------------------------|
| Missed detection or full occlusion | Occlusion hold; uncertainty grows    | Delayed transition (safe)           |
| False positive                     | Confidence gate and colour veto      | Suppressed                          |
| Red/yellow class swap              | Block transition; request reposition | <b>Cannot cause a wrong verdict</b> |
| Identity switch                    | Re-initialise track; reset debounce  | Delayed transition (safe)           |

**Evaluation and limits.** Detection is judged by precision, recall and mAP on a real held-out split, plus a dedicated **class-swap rate** for red/yellow, because standard mAP would not surface the failure that matters most. Nano detectors degrade on small or heavily truncated objects, and colour verification is itself lighting-dependent. Planned improvements include fiducial markers on the items for a deterministic identity check, and learned occlusion reasoning.

## 8. Orientation-Agnostic 3D Human Mesh Recovery

This chapter is the **research core** of the submission and the answer to optional requirement R9. Its status is in development for the pipeline and research-stage for the semi-supervised training components.

### 8.1 Why this is the crux

An Earth-trained pose network that meets an inverted astronaut swaps left and right limbs, mistakes global rotation for joint deformation, and loses tracking under occlusion. Each error corrupts the evidence the FSM depends on: a swapped hand gives the wrong hand-object distance, absorbed rotation gives wrong arm geometry, and collapsed tracking gives no evidence at all. This is not just lower accuracy. It is **structured, confident error**, which a deterministic validator cannot catch because its inputs are already wrong.

### 8.2 Pipeline

#### Frame to rack-frame joints

input frame (or AdaSpot RoI crop) → backbone features → 2D keypoints and 3D head (body parameters plus metric depth) → **orientation normalisation** → camera-space joints  $X_C$  → **transform**  $X_R = T(C \rightarrow R) \cdot X_C$  → rack-frame joints, split into the hand subset (for hand-object geometry) and the body mesh (for the Digital Twin)

Two stages make the system orientation-agnostic. Orientation normalisation removes orientation from the **representation**, and the rack-frame transform removes it from the **reference**. After the transform, "the left hand is 4 cm from the red box" means the same thing whether the astronaut is upright, sideways or inverted.

| Level                          | Content                                   | Used for                             |
|--------------------------------|-------------------------------------------|--------------------------------------|
| <b>2D keypoints</b>            | Joint positions in image pixels           | RoI centring, 2D overlay             |
| <b>Body parameters</b>         | Pose and shape of a parametric body model | Mesh generation, structural priors   |
| <b>3D joints, camera frame</b> | Metric position in camera frame C         | Input to the frame transform         |
| <b>3D joints, rack frame</b>   | Metric position in rack frame R           | <b>All downstream geometry</b>       |
| <b>Hand joints and mesh</b>    | Wrist and finger joints; surface vertices | Hand-object distance; twin rendering |

### 8.3 Training without microgravity labels

Annotated microgravity pose data is essentially unavailable, so we adopt the semi-supervised approach of PhysAstro-Pose (Cui et al., Sensors 2026). Three ideas are used:

- **Mean Teacher learning.** A teacher network (a moving average of the student) creates pseudo-labels on unlabelled frames, and the student trains on labelled plus pseudo-labelled data.

- **Structure-aware pseudo-label refinement (S-PLR).** Occlusion in a cluttered cabin produces anatomically impossible pseudo-labels. S-PLR models the skeleton as a graph and uses a graph convolutional network, trained on labelled data, to correct impossible configurations (a backwards elbow, an implausible limb length) before they reach the student. This works because microgravity postures are unusual but still anatomically constrained: joint limits do not change in orbit.
- **Canonical orientation constraint (COC).** A training-time constraint on body orientation that helps the network separate global rotation from joint deformation. It constrains in-plane orientation only, which is a stated limitation.

**Provenance.** The strong results reported for this method are the authors' results on their own dataset, not ours. We have adopted the method but not yet reproduced it. Our own accuracy and orientation-robustness figures will come from the evaluation protocol in a later chapter and will be reported as measured.

## 8.4 Astronaut position tracking and replay

Each processed frame yields the astronaut as a set of 3D landmarks (head, neck, shoulders, elbows, wrists, hands, hips, knees, ankles) in rack frame R, plus a reference point such as the pelvis and an orientation quaternion. Because everything is in R, position is expressed relative to the experiment rather than to the camera or the astronaut's own body. Over time this becomes a **trajectory**, a time-indexed record of position, joints, hands, orientation and confidence, exported as JSON or PKL. A recorded trajectory can be injected back into the Digital Twin to replay the movement **without rerunning perception**, which supports review, debugging, offline validation and dataset generation.

## 8.5 Failure handling and validation

| Failure                         | Response                                                            | Safety consequence                                  |
|---------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|
| Left/right limb swap            | Anatomical and temporal consistency check; keep previous assignment | Missing evidence delays the transition (safe)       |
| Depth collapse under truncation | Flag low confidence; widen position uncertainty                     | Distance threshold not met, so no transition (safe) |
| Hands fully occluded            | Declare degraded perception                                         | FSM holds state and issues no verdict               |
| More than one person in frame   | Track the subject nearest the rack; log ambiguity                   | Documented limitation                               |

**Validation** uses MPJPE and P-MPJPE on held-out data, an orientation-stratified test (upright, rotated, horizontal, inverted), and a separate **hand-joint error**, because whole-body error can hide hand-specific degradation and the hands drive every interaction decision. **Limits:** monocular metric depth depends on learned priors, and no real microgravity data exists for validation. **Future work:** temporal (sequence-level) reconstruction, hand-specific high-resolution recovery, and multi-view reconstruction if a second camera becomes available.

## 9. 2D-to-3D Reconstruction and the Rack Frame

### 9.1 The reconstruction chain

This is the second research component, and its central question is deliberately a question: **is monocular reconstruction accurate enough to support operational procedural decisions?** A dedicated experiment in the evaluation chapter answers it. The chain is:

#### From pixels to rack geometry

2D keypoints and object boxes → **lens undistortion** → metric depth  $d$  → back-projection with camera intrinsics  $K$  → 3D points in camera frame  $C$  → rigid transform  $T(C \rightarrow R)$  → 3D points in **rack frame  $R$**  → distances, containment and vectors → hand-object evidence

Back-projection uses the standard pinhole relation,  $X_C = d \cdot K^{-1} \cdot [u, v, 1]^T$ . Wide-angle payload cameras show strong radial distortion at the image edge, so keypoints and boxes are undistorted **before** back-projection. Ignoring this gives a position-dependent bias that would masquerade as depth error.

### 9.2 Camera calibration

| Step                              | What is done                                                                                                                                                                                                                                                                                     |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intrinsic</b>                  | A checkerboard or ChArUco target is captured at many poses across the working volume and image periphery, giving $K$ and the distortion coefficients. Target: RMS reprojection error of 0.5 px or less, with no systematic residual pattern. Repeated after any lens change or mechanical shock. |
| <b>Extrinsic (camera to rack)</b> | The rack origin and axes are defined physically. Fiducial markers (ChArUco or AprilTag) are placed at positions known in $R$ and detected in the image. The rigid transform is solved in closed form by the Kabsch/Umeyama method (SVD), which guarantees a proper rotation, not a reflection.   |
| <b>Error budget</b>               | The residual RMS is recorded as the calibration error, $\sigma_{\text{calib}}$ , and propagated into every downstream position uncertainty.                                                                                                                                                      |
| <b>Drift monitoring</b>           | A permanent fiducial stays in view at a known rack position. If its computed $R$ coordinates move, the system raises a calibration-drift alert and lowers confidence, instead of letting distance estimates degrade silently.                                                                    |

Because the camera is rigidly mounted to the payload structure, the transform is computed **once at installation** and is then static. Rack-frame tracking therefore costs one  $4 \times 4$  matrix multiply per point at runtime. Calibration tooling is in development.

### 9.3 The payload-rack coordinate frame $R$

Frame  $R$  is a right-handed Cartesian frame fixed to the rack, in metres. Its origin is a marked rack corner,  $+X$  runs along the horizontal working edge,  $+Y$  along the vertical structural edge, and  $+Z$  outward from the rack face.  **$+Y$  is "up" only relative to the rack.** In orbit there is no gravitational up, and the system never refers to one. The transform  $X_R = T(C \rightarrow R) \cdot X_C$ , with  $T$  in  $SE(3)$  (a rotation plus a translation), is applied uniformly to every point: joints, hand keypoints, object centroids and mesh vertices. Alongside it, a static rack model in  $R$  (workspace volume, container bounds, stowage

positions, camera pose) supports containment tests, workspace-exit detection, occlusion reasoning and twin rendering.

## 9.4 Why this delivers orientation-agnosticism

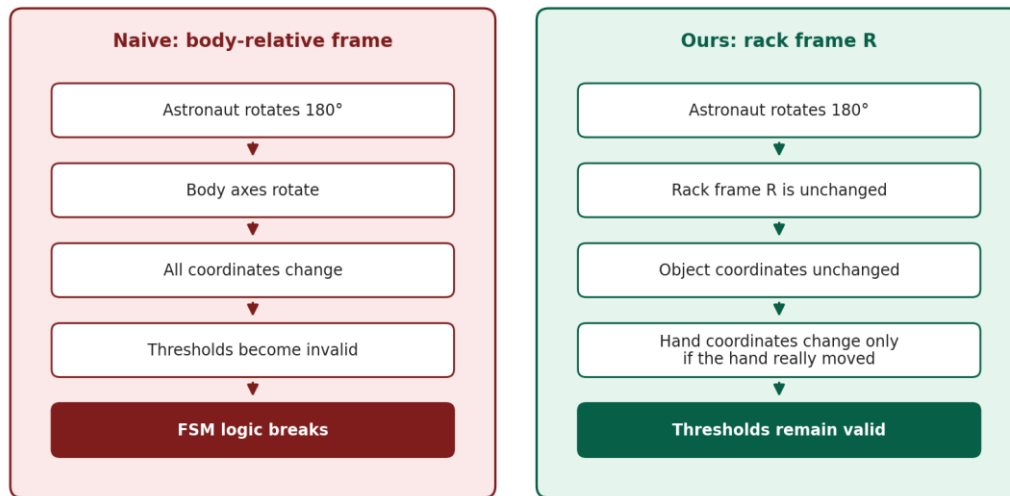


Figure 9.1: Body-relative frame versus rack frame when the astronaut rotates

The question "is the hand near the red box?" has an orientation-independent answer, because the hand and the box are located in the same static frame. The astronaut's body angle is not an input to that question at all. **Orientation-agnosticism is obtained by removing orientation from the question, not by training a model to be invariant to it.** The training work in the previous chapter is still needed to get accurate joints in the first place.

## 9.5 Depth estimation

| Depth source                                     | Basis                                                                     |
|--------------------------------------------------|---------------------------------------------------------------------------|
| <b>HMR metric head</b> (human joints)            | Learned body-scale prior; depth regressed in log space                    |
| <b>Known-size back-projection</b> (objects)      | Known physical width: $d \approx f \cdot W_{\text{real}} / w_{\text{px}}$ |
| <b>Contact anchoring</b> (objects in contact)    | Object depth is taken from the hand joint depth                           |
| <b>Rack-plane intersection</b> (objects at rest) | Viewing ray intersected with a known rack surface                         |
| <b>Temporal smoothing / UKF</b>                  | Recursive estimation across frames                                        |

The design combines these by confidence weighting. Contact-anchored depth dominates during grasp and extraction, so the system is **most accurate exactly when the FSM most needs it**. Depth is regressed in log space, which keeps it positive and gives a 10% error at 0.5 m and at 3 m a comparable loss.

**Stated without hedging**

This system **does not measure depth. It estimates depth.** Bounded accuracy, distance-dependent error and prior sensitivity are accepted and documented, and no operational threshold is trusted before the ground-truth experiment. A stereo pair or time-of-flight sensor would remove most of this uncertainty, but the problem statement specifies fixed payload cameras.

## 10. 3D Distance and Hand-Object Interaction

### 10.1 Estimated metric distance

Once all points live in rack frame R, the spatial relationships that form the **evidence** for procedural validation are computed. For two reconstructed points p and q the basic measure is the Euclidean distance  $d(p, q) = ||p - q||$ . This is a **reconstructed, calibrated metric distance estimate** derived from estimated depths and a calibrated transform. This document calls it an estimate throughout and never calls it exact. The evaluation described below decides what accuracy can honestly be claimed.

| Measure                           | Definition                                                  | Used for                                   |
|-----------------------------------|-------------------------------------------------------------|--------------------------------------------|
| <b>Hand-to-object</b>             | Minimum distance over hand joints and object surface points | Approach, contact, grasp                   |
| <b>Person-to-object</b>           | Body centroid or nearest joint to object                    | Workspace presence, skipped-step detection |
| <b>Object-to-object</b>           | Between object centroids                                    | Containment, placement                     |
| <b>Object-to-container bounds</b> | Signed distance to the container's 3D volume                | Extraction detection                       |
| <b>Lid displacement</b>           | Lid centroid movement along the opening axis in R           | Container-open transition                  |

Hand-to-object distance uses the minimum over hand joints and object **surface points**, not centroid-to-centroid distance. Otherwise a large box would register contact at half its diagonal.

### 10.2 Uncertainty of a distance

A distance between two uncertain points is itself uncertain. To first order,  $\sigma_d^2 \approx u^T(\Sigma_p + \Sigma_q)u$ , where u is the unit vector between the points and  $\Sigma_p, \Sigma_q$  are their positional covariances. The distance uncertainty is the combined positional uncertainty **projected onto the line between the points**. Distances along the optical axis, where depth is least certain, are therefore less reliable than distances across it, and this anisotropy is reported rather than averaged away. When both points come from the same HMR frame their depth errors are correlated, and a cross-covariance term is subtracted, which is why relative distance is better conditioned than absolute position.

### 10.3 Interaction events and thresholds

| Event             | Geometric condition                                                  | Temporal condition              |
|-------------------|----------------------------------------------------------------------|---------------------------------|
| <b>Approach</b>   | Hand-object distance falling, below about 20 cm                      | Monotone decrease over a window |
| <b>Contact</b>    | Distance at or below about 5 cm                                      | Persists for a minimum duration |
| <b>Grasp</b>      | Contact plus hand and object moving together (co-motion correlation) | Persists for a minimum duration |
| <b>Manipulate</b> | Grasp plus object displacement in R                                  | Continuous                      |
| <b>Extract</b>    | Grasp history plus object centroid leaving container bounds          | Persists for a minimum duration |

| Event   | Geometric condition                                                  | Temporal condition |
|---------|----------------------------------------------------------------------|--------------------|
| Release | Distance rises above contact threshold; object velocity goes to zero | Persists           |

The 20 cm and 5 cm values are **design starting points** taken from the team thesis, not validated thresholds. The threshold must be justified by measured error. If the estimator's error at operating range were 8 cm, a 5 cm contact threshold would be noise and would generate contact events from measurement error. So the adopted design rule is **threshold  $\geq k \times \text{RMSE of the distance estimate}$ , with  $k \geq 2$** , using the error measured at the operating range. The system also **refuses to start** with a threshold configuration its own measured accuracy cannot support, turning a silent accuracy problem into a loud configuration error.

## 10.4 Learned distance calibration

Because monocular reconstruction estimates rather than measures depth, geometric calculation and learned refinement are kept in two stages:

### Two-stage distance estimate

HMR and object detection  $\rightarrow$  3D reconstruction  $\rightarrow$  **geometric distance**  $d_{\text{geo}}$   $\rightarrow$  optional **calibration model**  $d_{\text{pred}} = f(X)$   $\rightarrow$  refined distance  $\rightarrow$  hand-object reasoning

The model input  $X$  may include the geometric distance, joint and object coordinates, estimated depth, HMR and detector confidence, object size, camera parameters, orientation and reconstruction uncertainty. The aim is **not to replace geometric reasoning** but to test whether a learned layer can remove systematic errors from monocular reconstruction and calibration. Keeping the stages separate lets the learned model be judged scientifically against the plain geometric estimate. In the same spirit, hand-object evidence is currently purely geometric and rule-based (explainable and deterministic), and any learned classifier for subtle cases such as a light touch would sit **upstream of the FSM as evidence**, never replacing the geometric checks.

## 10.5 Ground-truth evaluation

The learned model and the geometric estimate are both tested against physically measured distances in controlled hand-object configurations. For each observation we record the ground-truth distance, geometric distance, ML-predicted distance, absolute and percentage error, and confidence. The primary metrics are **MAE** (mean absolute error) and **RMSE** (root mean squared error). Tests vary distance, camera position, astronaut orientation, object size, partial occlusion, lighting and hand-object configuration. The resulting error distribution decides whether each spatial threshold is reliable enough for procedural validation, which is why this experiment is a mandatory gate before operational use.

| Failure                                          | Detection                                       | Response                         |
|--------------------------------------------------|-------------------------------------------------|----------------------------------|
| Depth collapse gives a spuriously small distance | Implausible for the box size; large uncertainty | Confidence gate blocks the event |
| Hand joint mislocalised                          | Anatomical or temporal inconsistency            | Discard that frame's evidence    |

| Failure                  | Detection                                    | Response                            |
|--------------------------|----------------------------------------------|-------------------------------------|
| Object depth unavailable | No size prior, no contact, not on rack plane | Mark distance unavailable; no event |

## 11. Sensing and Understanding Layers

Four components sit between raw geometry and the validator. None of them holds verdict authority: they add robustness, temporal understanding, efficiency and explanation.

| Component                    | Purpose                                                  | Status                                                   | Authority       |
|------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----------------|
| <b>IMU and fusion agents</b> | Inertial continuity between and during visual updates    | Planned                                                  | None            |
| <b>HAR agent</b>             | Recognise what the astronaut is doing over a time window | Rule-based version built; learned model under evaluation | None            |
| <b>AdaSpot Rol gating</b>    | Spend compute only where the interaction is              | Designed, not built                                      | None            |
| <b>Local VLM/LLM</b>         | Explanation and phrasing, entirely on-device             | Planned                                                  | None (advisory) |

### 11.1 IMU agent and visual-inertial fusion (planned)

**Status, stated plainly:** the IMU Agent and Fusion Agent are architecturally defined, with interface stubs and a specified filter, but the current prototype has **no physical IMU, driver or recorded inertial data**. The prototype runs vision-only, and the honest demo statement is that the visual path runs today and the inertial path is the next integration step.

The two sensors cover each other's weaknesses. Vision is drift-free but low-rate (10–15 Hz) and can vanish under occlusion. An IMU at 128 Hz is high-rate and always available but drifts, because integrating a small acceleration bias twice makes position error grow with the square of time. A constrained **unscented Kalman filter (UKF)** in rack frame R would fuse them, using the IMU to bridge short occlusions and to reject physically impossible visual jumps. The baseline is a **rack-mounted IMU**, so no crew-worn hardware or modified payload items are needed. Two microgravity points matter:

- **No gravity reference.** In free fall an accelerometer reads about zero, so attitude cannot be anchored to gravity and depends on gyroscope integration plus visual correction. This makes fusion more valuable, and it confirms that no sensor can supply a gravity-based "up".
- **Zero-velocity updates need rethinking.** Terrestrial methods rely on the gait cycle, which does not exist in orbit. The stationarity detector must be variance-based and tuned on recorded quiet data, which is an open research task.

Losing the IMU simply returns the system to vision-only mode, which is the implemented mode. The IMU is an enhancement, never a dependency.

### 11.2 HAR agent

Three questions are kept separate. HMR asks where the body is now. HOI asks what the hand-object relationship is now. **HAR asks what activity is occurring across a window of time.** HAR does not decide whether a step is complete: that is the FSM's job, made from HOI evidence and object state. HAR enriches the twin and informs guidance but is **not on the authority path**, so a probabilistic model can be used without importing its uncertainty into the verdict.

- **Structured inputs, not pixels.** The model sees Digital Twin features (joint and hand positions in R, object states, hand-object distances, contact flags, FSM state, confidences). It is therefore small, interpretable, and robust to lighting and background changes.
- **Model choice is not finalised.** Candidates are sliding-window rules (the current behaviour), a temporal CNN, a lightweight recurrent network and a small transformer. The choice will follow measured accuracy and latency on our hardware.
- **Benchmark classes** include idle, approaching, opening, reaching into container, grasping and extracting red or yellow, placing, and left workspace. **Uncertain is a first-class output.** A system that always emits a confident label lies when it does not know, and uncertain never triggers a transition.
- **Stability and validation.** A label must hold for a set time before it changes, which prevents flicker. The confusion matrix matters more than accuracy, because the dangerous error is grasping-red versus grasping-yellow.

### 11.3 AdaSpot: region-of-interest processing

Most pixels in a rack-camera frame are static background, so running full-resolution HMR and detection on every frame wastes the thermal budget. AdaSpot uses **two resolutions**: a cheap low-resolution context pass runs on every frame and maintains a saliency map, and the expensive high-capacity detection and HMR run only on a cropped region of interest, mapped back to full-frame coordinates. Saliency combines motion, an interaction prior (regions near hands and known objects, even if nothing moves) and a procedural prior from the FSM state. That last cue is a **priority weight, never a mask**, because masking unexpected regions would blind the system to the out-of-order actions it must catch. The published method reports up to 60% fewer GFLOPs, but that is the paper's result, not ours. Our own saving will be measured in the ablation study.

### 11.4 Local VLM/LLM and its restrictions

The VLM/LLM runs entirely on the device: no cloud API, no external service, no internet, quantised weights held locally. This is requirement D1, and it can be demonstrated with the network disabled. Its permitted roles all sit **downstream of a decision the deterministic layer has already made**: describing the situation, summarising steps, explaining a verdict, phrasing guidance and writing an end-of-session report.

| It may never                                         | Why                                                             |
|------------------------------------------------------|-----------------------------------------------------------------|
| Declare a step complete or the experiment successful | A fluent, unverified assertion                                  |
| Override, suppress or downgrade an anomaly           | A hallucinated "this looks fine" would defeat the safety design |
| Modify twin state or trigger a transition            | Authority belongs only to the FSM                               |
| Be the sole source of a voice alert                  | Alerts must trace to deterministic evidence                     |

The prohibition is **enforced in the architecture, not in a prompt**, because prompts are not a security boundary. It has five layers: the VLM's output type accepts no function that mutates twin or FSM state (a write does not compile); it receives a read-only snapshot; the FSM's transition function has no parameter for language-model output; advisory text is logged in a separate field the validator never

reads; and on the target hardware it runs in a separate fault domain. The worst case of a hallucination is then **a wrong sentence on a screen beside a correct verdict**, visibly inconsistent and marked advisory. The entire generative layer can fail without affecting any requirement. Candidate models are small quantised instruction-tuned models (about 1–4 billion parameters); **none has been selected**.

## 12. Deterministic Validation Core

### 12.1 Why the verdict must be deterministic

A procedural validator needs four properties that no probabilistic model provides: **determinism** (same evidence, same verdict), **explainability** (the verdict breaks into named conditions), **auditability** (reconstructible from the log months later) and **bounded behaviour** (finite, enumerable states, since you can only certify what you can enumerate). A finite-state machine gives all four by construction. Neural networks are not bad; authority simply needs properties they lack, so each tool is used where it is strong.

The validator is a deterministic automaton  $M = (S, \Sigma, \delta, s_0, F, E)$  with finite states, a finite alphabet of twin-derived evidence events, a **total** transition function, an initial state, accepting states and error states. Totality means no undefined input: an unexpected event maps to an explicit outcome, usually "flag as out-of-order and stay put". Because it consumes only twin state, it is tested with synthetic sequences (every transition, anomaly and recovery, plus flicker at thresholds) and needs no camera, model or human. Status: implemented.

### 12.2 Benchmark state model

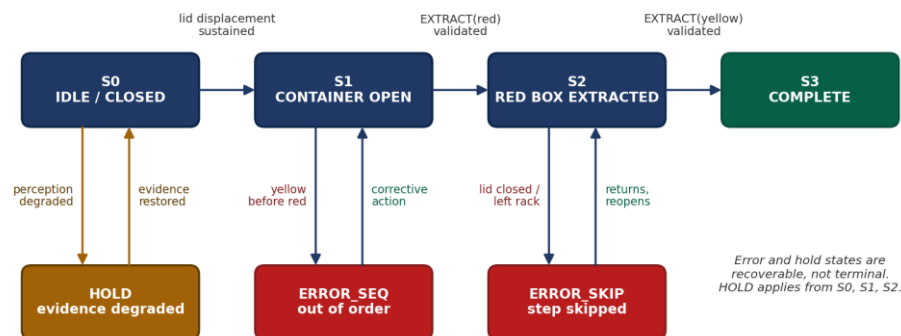


Figure 12.1: FSM for the red/yellow box experiment

| State                            | Advances when                                                                                                                                 | Voice message                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>S0</b> idle, container closed | Lid moves beyond about 15 cm along its opening axis, sustained. Any extraction before this gives ERROR_SEQ.                                   | "Please open the primary container box."    |
| <b>S1</b> container open         | Red box extraction is validated (grasp, co-motion, leaves container bounds, colour verified, continuous track). Yellow first gives ERROR_SEQ. | "Next step: please extract the red box."    |
| <b>S2</b> red box extracted      | Yellow box extraction is validated. Lid re-closed or astronaut away too long gives ERROR_SKIP.                                                | "Next step: please extract the yellow box." |
| <b>S3</b> complete               | Accepting state; a session summary is generated.                                                                                              | "Experiment successfully complete."         |

The 15 cm lid value is a design starting point, to be recalibrated against measured error. Error states are **recoverable**: an astronaut who reaches for the wrong box, hears the alert and corrects returns to the right state. A validator that latched into failure on the first mistake would be useless.

## 12.3 Temporal debouncing

A guard true on one frame does not commit a transition. A counter grows while the guard holds, resets when it fails, and commits only after enough persistence:

- **Specified in seconds, not frames.** Transitions need 0.5 s and anomalies 0.3 s. The inherited "15 frames" was 0.5 s at 30 FPS but 1.0–1.5 s at our real 10–15 FPS, delaying alerts past the useful window. Seconds also stay stable under thermal throttling.
- **Asymmetric on purpose.** Slower to praise, faster to warn: a late warning may arrive after the wrong box is out.
- **Gaps reset the counter.** Evidence before and after a gap is never joined into one transition.

## 12.4 Anomaly classes and the evidence record

| Class                   | Meaning                                                                    | Corrective instruction                   |
|-------------------------|----------------------------------------------------------------------------|------------------------------------------|
| <b>ERROR_SEQ</b>        | An action valid later in the procedure is done now                         | Names the required action first          |
| <b>ERROR_SKIP</b>       | A required step is abandoned or reversed (lid re-closed, astronaut leaves) | Names the outstanding step               |
| <b>HOLD</b>             | Perception too degraded to judge; advisory, not an error                   | "Tracking degraded, please reposition"   |
| <b>AMBIGUOUS_OBJECT</b> | Colour check disagrees with the detector                                   | Asks the astronaut to separate the boxes |

The split matters because the corrective instruction differs: out-of-order says do this other thing first, skip says go back and finish what you left. Every committed transition and anomaly also writes an **evidence record** (guard, measured values, uncertainty, duration, debounce counts), so a reviewer weeks later sees not just the decision but the measurements behind it.

## 12.5 Graph-guided validation (planned)

Real procedures are not always linear. If three samples may be prepared in any order, a linear FSM would flag valid orderings as errors, and a validator that raises false alarms on correct work gets ignored. The planned extension models the procedure as a **directed graph**: nodes are experiment states, edges are permitted transitions carrying the same evidence guards, with support for branches, convergence, optional nodes and bounded repeats. **The graph computes what is allowed, and the FSM enforces what was done.** An event is then valid, out-of-order (belongs elsewhere in the procedure), skipped (bypasses a mandatory node) or illegal (in no part of the procedure).

Skip detection uses **dominator analysis**: a node is mandatory if every path to completion passes through it. This runs once at protocol load, so runtime cost is a set lookup. The protocol is also statically checked at load, like type-checking a program (reachable terminals, guarded edges, bounded repeats, thresholds above the noise floor). **Determinism is preserved**, since the same evidence always takes the same path. With several valid next steps, guidance says "any order" and never implies a constraint that does not exist.

## 12.6 Reasoning agent

The Reasoning Agent turns a **validated** state into something the astronaut can act on, in two tiers. **Tier 1 is deterministic:** every state has a guidance template written at protocol-authoring time, so correct guidance always exists and requirement R2 is met without any model. **Tier 2 is optional:** the local VLM may rephrase, but only after a consistency check against twin facts; otherwise Tier 1 is used unchanged. Messages are prioritised, rate-limited so the system is not muted for repeating itself, and alerts **pre-empt** speech in progress. It reads only post-verdict fields and writes only advisory ones.

## 13. Outputs and Interfaces

**PURPOSE** — everything through which the system is seen, heard, recorded, and downlinked: overlay video, voice alerts, the mission-control GUI, structured telemetry, and local/streamed video. None of these components holds verdict authority (§11); they present it.

*Status: [A] implemented — GUI, offline TTS, JSONL telemetry, local recording, and HTTP streaming all operate in the current prototype.*

### 13.1 Monitoring Agent and overlay rendering

The Monitoring Agent composes and serves the GUI, renders 2D/3D overlays, drives TTS playback, streams video, and writes telemetry (R4, R5, R6, R7, R9). The overlay is not decoration: it renders bounding boxes with class/confidence/colour-verification status, projected 3D joints with hand joints highlighted, the container's 3D bounds, the live d\_HOI value and its uncertainty as a hand-object line, and the AdaSpot RoI when active — rendering the decision geometry, not just detections, so the system's reasoning is visible for debugging and demonstration alike. The agent fails loudly: on audio loss it escalates to a full-screen visual banner rather than dropping an alert silently; on disk-full it stops recording (large, replaceable) before telemetry (small, irreplaceable). Resource-exhaustion priority is fixed: telemetry > validation continuity > audio > GUI > streaming > recording.

### 13.2 Offline text-to-speech

Voice (R3) is used because an astronaut's eyes are already committed to a manipulation, and audio needs no facing direction in microgravity. TTS runs fully offline (D1), with low, bounded latency, intelligibility under cabin noise, a consistent voice, and pre-emptible priority. The key latency trick is phrase caching: every state's guidance and every anomaly's warning is known at protocol-load time, so all fixed phrases are synthesised once at startup and cached; at runtime the common case is near-zero-latency playback of a pre-rendered clip, and only dynamic VLM narrative — never on the alert path — needs runtime synthesis. Safety measures: warnings carry a distinct tone so they are recognisable before the words are parsed; routine guidance is never delivered in an alarm register; audio failure escalates to visual alerting (§12.1) rather than failing silently; and an already-acknowledged anomaly is not re-spoken. If the engine fails to initialise or the device is lost mid-session, the system falls back to visual-only alerting and declares the degradation — R3 is reported as degraded, never hidden.

### 13.3 Graphical user interface

The GUI is a zero-dependency, vanilla HTML/CSS/JS single-page application served locally by the same process that runs the pipeline — no build step, no CDN, no internet, trivially auditable. Live video is consumed as an MJPEG <img> stream with a /snapshot polling fallback; telemetry is polled at 25 Hz (deliberately faster than the 10–15 Hz inference rate, so the display never lags state by more than one poll interval — not a claim that inference itself runs at 25 Hz).

| Panel                  | Contents                                                     |
|------------------------|--------------------------------------------------------------|
| Live video + overlays  | MJPEG stream with 2D/3D overlays                             |
| Step verdict / actions | NOMINAL / ERROR / HOLD badge, current + next required action |
| Procedure checklist    | Per-step completed / active / pending                        |
| 3D Digital Twin        | Rack-frame rendering of astronaut and objects                |

| Panel          | Contents                                                                             |
|----------------|--------------------------------------------------------------------------------------|
| Diagnostics    | f_inf, dropped frames, CPU/GPU, thermal, disk, TTS, calibration                      |
| Event timeline | Scrolling telemetry event feed                                                       |
| Advisory (VLM) | Non-authoritative commentary, visually de-emphasised — a safety feature, not styling |

A native Qt console remains a candidate for the flight HMI where display-terminal overhead must be minimised; this document records the browser SPA as the implemented interface, not both as existing.

### 13.4 JSONL telemetry

Telemetry (R4) is written as JSON Lines — one self-describing object per line — chosen for being append-only, streamable, crash-safe (a truncated file loses at most its last line), and readable with ordinary text tools. Events span eight categories: session, state, detection, interaction (hoi\_contact/grasp/extract/release), anomaly (error\_seq, error\_skip, hold\_entered...), guidance, diagnostics, and inference metadata. Every record carries a common envelope (timestamp, monotonic time, session/experiment IDs, a monotonically increasing seq that exposes gaps, event, FSM state, status, message), plus category-specific fields such as distance and its uncertainty or the full evidence dictionary for a transition or anomaly.

Because every transition is logged with the evidence that justified it and the FSM is deterministic (§11.1), a recorded session can be replayed through the validator to reproduce its verdict sequence — the basis for regression testing, post-hoc PI review, and anomaly investigation. On sizing: an earlier team claim of a "3,000,000:1 compression ratio" against raw uncompressed video is not adopted here — telemetry encodes events, not the video, and the comparison baseline is not what the system actually stores. What is stated instead, and is measurable, is that telemetry size scales with the number of events, not with duration or resolution, keeping a session's log on the order of tens of kilobytes — event-driven logging, periodic (not per-frame) health sampling, and evidence attached only to transitions/anomalies are what keep it there.

### 13.5 Local recording and network streaming

R5 (stream to a specified IP) and R6 (store locally) are served as two independent paths both drawn from the same ring buffer, deliberately not derived from one another: a viewer connecting or disconnecting cannot perturb the recording, and a disk-full condition cannot interrupt the stream. The stream stays on the station LAN — it is not the downlink path; §12.4's telemetry is.

| Aspect            | Path A — local recording                                                                                                 | Path B — network streaming                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Codec / protocol  | H.264, full rate (30 FPS) even though inference is decimated — discarding unseen frames would destroy re-analysable data | MJPEG/HTTP today; RTSP targeted for the flight build                    |
| Why               | Scientific record, retrospective re-analysis, training data, dispute resolution                                          | Live viewing on crew tablet / second terminal, no client install needed |
| Failure isolation | Disk full → stop recording, preserve telemetry, alarm                                                                    | No clients → idle; LAN saturation → reduce rate/resolution              |

## 14. Dataset, Training and Edge Deployment

**PURPOSE** — build the custom corpus R8 demands, specify what each model is trained on, and define the hardware/software stack the trained pipeline runs on. No public dataset covers this container, these boxes, microgravity postures, or procedure-specific error sequences, so the corpus is necessarily built in-house.

*Status: [B] synthetic pipeline and real capture in progress; [B]/[D] no model finalised or benchmarked on our own training run; [C] edge deployment target (Jetson Orin NX, TensorRT/INT8).*

### 14.1 Synthetic generation and domain randomisation

Five sources are combined: a real physical mock-up (ground truth / sim-to-real anchor), synthetic rendering (bulk volume with exact, free labels), domain-randomised synthetic (robustness), public pose corpora (backbone pre-training only), and augmented real capture (bridging). The renderer knows every object pose, joint position and interaction state exactly, at no annotation cost. Randomised factors — body orientation (full range, upright to inverted), NBP-consistent posture, lighting, camera pose/noise, occlusion, object/background variation and distractors — each target a specific failure mode from §3; orientation randomisation is the primary mechanism that prevents an upright "gravity prior" from forming in training, not an incidental augmentation.

Sim-to-real is closed by mixed-batch training (every batch mixes synthetic and real), semi-supervised Mean-Teacher adaptation on unlabelled real frames, progressive fine-tuning, and a real-only test split that measures the gap rather than hiding it. Stated honestly: synthetic data does not reproduce true microgravity dynamics — how a body settles in free fall, how unsecured objects drift — only appearance; closing that residual gap needs parabolic-flight or analogue data (roadmap). An earlier "millions of frames" claim is not adopted; actual corpus size is reported once generated.

### 14.2 Sequence classes, labels and splits

Recorded sequences span seven classes — correct, out-of-order, skipped, corrected, near-miss, degraded, ambiguous — each mapped to a specific validator behaviour (ERROR\_SEQ, ERROR\_SKIP, HOLD, AMBIGUOUS\_OBJECT). Near-miss (hand approaches the wrong box without grasping) is the most often omitted and the most important: without it, false-alert rate — the metric that determines whether crew keep trusting the system — cannot be measured. Labels are collected per target: 2D/3D detection boxes and pose, hand keypoints, per-frame HOI state (APPROACH...RELEASE), per-window activity class, ground-truth FSM state and transition/anomaly frame indices, and ground-truth metric distances.

Splits are ~70/15/15 train/validation/test, partitioned by session and subject — never by random frame, which would leak near-duplicate frames across splits and inflate scores that then collapse in deployment. The test split is real-data-only, so reported numbers measure the sim-to-real gap rather than concealing it.

### 14.3 Model inventory (M1–M6)

| #  | Model                  | Approach / trained on                                                                                                                                                      |
|----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 | Object detector        | YOLOv8-nano; synthetic (bulk) + real mock-up (anchor); test split real-only; INT8 post-training quantisation validated against FP32 mAP loss                               |
| M2 | 3D Human Mesh Recovery | Backbone + keypoint/3D/orientation heads; semi-supervised Mean-Teacher; public pose pre-train → synthetic microgravity → unlabelled real; full-range rotation augmentation |

| #  | Model         | Approach / trained on                                                                                                                                                                                 |
|----|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M3 | HOI evidence  | Geometric/rule-based (distance, persistence, co-motion) — explainable, no training data needed; a learned classifier over the same features is a possible future refinement, upstream of the FSM only |
| M4 | Temporal HAR  | Sliding window of structured geometric features (not pixels); candidates: TCN, lightweight RNN, small transformer; selected by measured accuracy/latency on target hardware, not published numbers    |
| M5 | Local VLM/LLM | Off-the-shelf small quantised instruction-tuned model; reads a structured twin snapshot and rephrases — never decides; output consistency-checked before display                                      |
| M6 | Offline TTS   | Off-the-shelf offline neural TTS, used as supplied, optimised only via startup phrase caching                                                                                                         |

Training discipline: every run logs dataset version, split hashes, seed, hyperparameters and commit; the test split is touched once, at the end; and any reported accuracy must state model/dataset/split/hardware/seed-variance together — a single number without those is not treated as a result.

## 14.4 Edge hardware and software stack

| Layer                    | Technology                                                                                                          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| Language / orchestration | Python 3 — agents, pipeline, blackboard (src/core/shared_memory.py)                                                 |
| Perception (M1–M3)       | YOLOv8-nano; HMR backbone + heads; geometric HOI rules                                                              |
| Saliency gating          | AdaSpot-style dual-resolution RoI                                                                                   |
| Geometry / calibration   | Camera model, camera→rack transform, covariance propagation; planar intrinsic + fiducial (Kabsch/Umeyama) extrinsic |
| Inertial / fusion        | 6-DoF IMU + ZUPT; constrained UKF                                                                                   |
| Validation / reasoning   | Deterministic FSM + graph validator; Tier-1 templates + Tier-2 local VLM                                            |
| Speech / GUI             | Offline neural TTS + phrase cache; vanilla HTML/CSS/JS SPA served locally                                           |
| Video / telemetry        | H.264 record; MJPEG/HTTP (RTSP target) stream; append-only JSON Lines                                               |
| Edge inference target    | TensorRT / INT8 on Jetson Orin NX                                                                                   |
| Fault isolation          | Deterministic co-processor partition separating validation from perception                                          |

## 15. Benchmark Walkthrough, Errors and Safety

**PURPOSE** — trace one frame end to end, walk the reference procedure and its failure modes stage by stage, catalogue every anomaly and fault the system anticipates, and state in one place why the resulting verdict can be trusted.

*Status: [A] nominal walkthrough, anomaly taxonomy, and safety hierarchy match the implemented FSM; scenario I (graph branching) and several fault detectors are [C] planned, specified at implementation depth.*

### 15.1 End-to-end execution — one frame, 24 steps

Acquisition (1–5): camera capture → timestamp → ring buffer, from which recording, streaming and the inference scheduler independently fork → pre-processing. Perception (6–10): AdaSpot RoI gating → YOLO detection → colour verification/tracking → HMR reconstruction (2D/3D joints, mesh) → hand-joint extraction. Geometry (11–13): per-joint/object depth → back-projection to camera frame → transform to rack frame R, all with covariance carried forward. Sensing fusion (14–15, planned): IMU ingestion and bias correction → visual-inertial UKF fusion. Understanding (16–17): HOI distance matrix → temporal HAR window and activity label (or UNCERTAIN). State and decision (18–22): Digital Twin update and immutable snapshot → graph allowed-set evaluation (planned) → FSM guard/debounce evaluation issuing state, verdict, anomaly class and evidence record → Reasoning Agent guidance. Output (23–24): GUI/TTS rendering → JSONL event append, video recorded and streamed. The chain is strictly one-directional up to the verdict (step 21); nothing downstream of it can rewrite an upstream fact.

### 15.2 Benchmark walkthrough — open, extract red, extract yellow

| Stage                 | Perception / HOI                                                                                          | FSM                                               | Voice                                                          |
|-----------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------|
| 0 · Init              | container + lid (closed) detected; astronaut ~0.6 m away; d(hand,container) ≈0.58 m, HOI = NONE           | S0, next = open_container                         | "System initialised. Please open the primary container box."   |
| 1 · Open              | lid tracked through APPROACH → CONTACT (0.31s) → GRASP (p=0.89) → MANIPULATE; lid displacement 0.183 m    | guard >0.15 m sustained 0.52 s ≥ τ → commit S0→S1 | "Container opened. Next step: please extract the red box."     |
| 2 · Red box           | d(hand,red)→0.031 m; GRASP(p=0.91, 0.83 s)→EXTRACT; all seven \$HOI conditions pass incl. colour_verified | commit S1→S2                                      | "Red box extracted. Next step: please extract the yellow box." |
| 3 · Yellow box / done | same progression on yellow_box; both boxes EXTRACTED, container open+empty                                | commit S2→S3 (terminal)                           | "Yellow box extracted. Experiment successfully complete."      |

### 15.3 Failure-scenario walkthroughs (A–I)

| # | Scenario                                          | Verdict                | Key point                                                                                                                                             |
|---|---------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | Yellow grasped before red (in S1)                 | ERROR_SEQ              | Guard matches an edge not reachable from here; shorter anomaly debounce (0.3 s) so the warning arrives while the box is still in hand; state stays S1 |
| B | Container re-closed with yellow outstanding       | ERROR_SKIP             | Removing the enabling condition with a mandatory step outstanding is treated as a skip, not silently accepted                                         |
| C | Astronaut leaves workspace mid-procedure          | ERROR_SKIP (departure) | Distinguished from occlusion by the body being tracked and outside, not untracked; alert is queued and delivered on re-entry, not into an empty room  |
| D | Temporary occlusion (~0.6 s)                      | HOLD                   | Staleness exceeds the transition gate — no transition, no anomaly. Occlusion is never conflated with an error                                         |
| E | Detector loses the target (8 frames, no occluder) | HOLD → object_lost     | Extraction is never inferred from mere disappearance — the object must be positively detected outside the container                                   |

| # | Scenario                                                        | Verdict                   | Key point                                                                                                                     |
|---|-----------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| F | HMR confidence drops (extreme pose / self-occlusion)            | HOLD                      | Confidence gate fails, covariance widens; the same code path handles nominal and degraded operation — only the numbers change |
| G | Hand brushes the correct box without grasping                   | No transition, no anomaly | Co-motion $p=0.12 \ll \text{threshold}$ — the near-miss case that determines whether the system is usable in practice         |
| H | Boxes adjacent; colour verification disagrees                   | AMBIGUOUS_OBJECT          | The system asks rather than guesses — a confident wrong guess would be the worst available outcome                            |
| I | Branching protocol, any of several valid next actions (planned) | VALID, no anomaly         | Without the graph, legitimate alternative orderings would be flagged as errors and crew would stop trusting alerts            |

## 15.4 Anomaly taxonomy

Procedural anomalies (ERROR\_SEQ, ERROR\_SKIP, ERROR\_ILLEGAL — the astronaut did something inconsistent with the procedure) are kept strictly distinct from system anomalies (HOLD, PERCEPTION\_DEGRADED, CAMERA\_FAULT, CALIBRATION\_DRIFT, AMBIGUOUS\_OBJECT — the system cannot perceive reliably), because telling an astronaut they erred when the camera was actually blocked destroys trust in every later alert. Procedural anomalies carry a warning register and imply the procedure is at fault; system anomalies carry an advisory register, hold state without attributing error, and imply the system is at fault. An anomaly's lifecycle passes through a debounced CANDIDATE stage (filters transient noise) and, if unacknowledged, a bounded ESCALATED stage — bounded because an alarm that never stops is an alarm that gets disabled.

## 15.5 Fault-handling contract and catalogue

Every fault obeys four rules: detected, never assumed absent (an explicit detector exists for each); announced, never silent (telemetry plus, where relevant, an operator-visible indication); degraded, never wrong (uncertainty is answered by withholding a verdict, never guessing one — the load-bearing rule); and recoverable where possible, with the recovery logged. Twenty-four credible faults are catalogued across five groups:

| Group                     | Representative faults                                                                                                 | Typical fallback                                                                                                                  |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Acquisition               | Camera failure, dropped frames, sensor desync, clock anomaly                                                          | Freeze twin / suspend transitions; reset debounce counters; reject non-monotonic samples                                          |
| Perception                | Detection failure, false detection, pose-estimation failure, occlusion, excess depth uncertainty, HMR/IMU drift       | Reject or widen-uncertainty the affected evidence; re-acquire on recovery; never inferred from absence                            |
| Generative layer          | VLM hallucination, VLM unavailable/OOM                                                                                | Suppress advisory / Tier-1 templates only — no safety consequence, since the layer holds no authority                             |
| Resource                  | Disk full, telemetry write failure, GPU overload, memory exhaustion, thermal throttling                               | Priority-ordered shedding: stop recording before telemetry; unload VLM before overlays; proactively throttle before hardware does |
| Configuration / integrity | Invalid transition attempt, protocol definition error, calibration drift, agent unresponsive, guard below noise floor | Rejected by a total transition function; load-time static validation; continuous fiducial monitoring; startup refusal             |

Telemetry write failure and disk-full carry the highest safety consequence (the scientific record is at risk) and are mitigated by the resource priority telemetry > validation continuity > audio > GUI > streaming > recording; audio/TTS failure escalates to visual alerting rather than failing silently.

## 15.6 Safety by design

Six layers narrow what passes to a verdict — neural perception (probabilistic, high capability, not trustworthy alone) → thresholded/confidence-gated evidence → the structured, staleness-aware

Digital Twin → deterministic validation (finite, enumerable, debounced, replayable) → an authorised transition committed with its justifying evidence → guidance/alert. The generative layer (VLM/Reasoning) reads the twin read-only and writes text-only output to the alert layer; it has no path into validation. This is enforced at five levels — type (advisory output has no state-mutating function that accepts it), interface (read-only snapshot), data-flow (the FSM's transition signature admits only enumerated evidence fields), telemetry (advisory text sits in a field the validator never reads), and, on target hardware, partition (separate fault domains) — because a policy is a request and a type system is a constraint.

The fail-safe principle, stated precisely: when the system cannot establish the required evidence, it does not issue a verdict — not a low-confidence one, not a best guess, but an announced HOLD. This biases errors toward omission rather than commission: the system may occasionally fail to confirm a correctly performed step (mildly annoying), but it will not confirm a step that was not performed and will not accuse an astronaut of an error they did not make — the right asymmetry for a procedural-integrity system, since a false confirmation corrupts the record silently while a withheld verdict is visible, explicable and recoverable.

Stated honestly, this architecture does not protect against everything: systematically wrong perception (a confidently mislocated hand validates wrong evidence correctly — mitigated only partially by confidence gating, colour verification and multi-condition checks), calibration error (biases all distances coherently — mitigated by continuous fiducial monitoring), a wrong protocol specification (validated perfectly and wrongly — mitigated by load-time static checks and human review), and unmodelled procedure variation (flagged as an error until encoded in the graph). The deterministic validator makes the decision layer trustworthy; it does not make the perception layer correct — which is precisely why confidence gating, the evaluation plan (§15), and the willingness to withhold a verdict all exist.

## 16. Evaluation Plan

**PURPOSE** — define, per subsystem, what is measured, on what data, and against what acceptance target. No measured results appear here: every table is a specification of an experiment to be run. Cells marked TBM are to-be-measured; a future revision replaces them with (M)-tagged values carrying model/dataset/split/hardware/seed-variance provenance (§13.3).

*Status: [C] — plans specified at implementation depth; the 2D-to-3D experiment (§15.2) is the highest-priority outstanding measurement, since every operational threshold in the system depends on its outcome.*

### 16.1 Per-module metrics

| Module              | Key metrics                                                                                                                                                                    | Target                                                                                                                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detection (M1)      | Precision, recall, mAP@0.5, red↔yellow class-swap rate, latency, INT8-vs-FP32 delta                                                                                            | P/R ≥0.90; mAP@0.5 ≥0.85; swap-rate ≤0.5% (tightest — the only failure that is confidently wrong, not missing); latency ≤35 ms                                                                                   |
| HMR (M2)            | MPJPE, P-MPJPE, hand-joint MPJPE, depth MAE, orientation-stratified error, latency                                                                                             | Hand-joint MPJPE is operationally binding; inverted-vs-upright degradation ≤25%; latency ≤55 ms. MPJPE vs P-MPJPE separates wrong-pose from wrong-orientation errors                                             |
| Distance            | MAE, RMSE, % error, bias, anisotropy, calibration contribution                                                                                                                 | MAE ≤2 cm at 1 m; % error ≤5%;  bias  ≤1 cm (systematic offset is recalibratable, reported separately from scatter)                                                                                              |
| HOI (M3)            | Event precision/recall/F1, detection latency, false-event rate on near-miss data                                                                                               | P/R/F1 ≥0.95; latency ≤1.0 s; false-event rate ≤2% on the near-miss class specifically                                                                                                                           |
| HAR (M4)            | Accuracy, per-class P/R/F1, confusion matrix (red↔yellow inspected), UNCERTAIN rate, latency                                                                                   | Accuracy ≥0.90; latency ≤20 ms; a suspiciously low UNCERTAIN rate flags overconfidence                                                                                                                           |
| Sequence validation | Correct-sequence rate, false-alert rate, missed-error rate, transition accuracy/latency, anomaly-to-alert latency, anomaly classification accuracy, recovery rate, determinism | The R1/R3-level metrics a jury should ask about: correct-sequence ≥98%; false-alerts ≤0.05/session; anomaly-to-alert ≤1.5 s; determinism = 100% (an architectural guarantee to verify, not a statistical target) |

System-level (not detailed here): end-to-end latency ≤200 ms, inference 10–15 FPS, capture/record at 30 FPS, ≥2 h thermal/memory stability, startup ≤30 s, and telemetry size reported in KB (replacing the earlier unverified compression-ratio claim, §12.4). Protocol discipline: ≥10 sessions per scenario, ≥3 subjects, nominal + degraded conditions, dev/target hardware never merged, test split touched once.

### 16.2 The 2D-to-3D evaluation experiment

This experiment answers the central research question directly: does monocular HMR-based 3D reconstruction give geometry accurate enough to justify a 5 cm contact threshold — not merely to produce a plausible mesh? Its output feeds §16.4's rule (threshold ≥ 2×RMSE), which sets every contact/grasp threshold, which sets every HOI event, which sets every FSM guard — so an unmeasured pipeline means every operational threshold in the system is presently an intuition, not a number.

Setup: the deployment camera at deployment mounting/resolution, full intrinsic+extrinsic calibration with residuals recorded, a laser/tape ground-truth instrument (≤2 mm uncertainty), the actual physical props, ≥3 subjects of differing build, ≥10 repetitions per condition, every trial logged to telemetry with its ground-truth value as a field. Distance sweep spans 0.05–2.00 m (contact regime through module-scale) at seven fixed points, reporting mean estimate, absolute/percentage error, scatter, and — critically — whether the system's own reported uncertainty  $\sigma_d$  matches observed scatter (an unvalidated  $\sigma_d$  is decoration, not a safeguard).

| Sweep            | Conditions                                                                                                                              | Acceptance                                                                                                                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orientation (R9) | Upright 0°, rotated 45°, horizontal 90°, inverted 180° ( $\pm 20^\circ$ bands)                                                          | Inverted distance MAE $\leq 1.25 \times$ upright MAE — failure means the orientation-agnosticism claim is reported as unsupported, not quietly hidden by restricting demos to upright poses |
| Degradation      | Occlusion 0–75%; lighting (nominal/low/harsh/coloured/specular); camera angle; object size; hand-object closing distance; subject build | Distance MAE, event-detection rate, colour-verification failure rate, HOLD rate, anisotropy — each reported per condition                                                                   |

Terrestrial inversion is a stated approximation (physically inverted/horizontal relative to the rack) that reproduces the visual condition but not the biomechanical one (NBP, unloaded musculature); true microgravity validation is roadmap Phase 9.

### 16.3 Ablation study

Thirteen configurations isolate what each component contributes, from detection-only (A) through added 2D pose (B), 3D HMR (C), geometric HOI (D), temporal HAR (E), IMU/UKF fusion (G), graph-guided validation (I), VLM advisory (J), RoI gating (K), quantisation level (L), to the seconds-vs-frames debounce correction (M) — each measured on accuracy, robustness under §15.2's degradation sweeps, latency, false-alert rate, sequence-validation rate, distance error, and compute cost. Two evaluation-condition rules are enforced: fusion (G) must be evaluated under degraded, not only clean, conditions — it helps exactly where vision is worst, so a clean-only test would wrongly conclude it is unnecessary; and the graph validator (I) must be evaluated on a branching protocol, since on the linear benchmark it is equivalent to the FSM by construction. A component that does not measurably help is removed — complexity that survives only because it is in the design document is technical debt on a thermal budget. Status: [C] planned; no results exist or are invented here.

### 16.4 Requirement traceability matrix

| ID        | Requirement                                                 | Primary component(s)                       |
|-----------|-------------------------------------------------------------|--------------------------------------------|
| R1        | Continuously track the experiment sequence from local video | acquisition/perception/HMR, HOI, FSM       |
| R2        | Suggest the next step at start and after each step          | next expected, Reasoning Agent             |
| R3        | Voice alert on skipped/out-of-sequence step                 | anomaly classes, offline TTS               |
| R4        | Timestamped structured lightweight log of steps/outcomes    | JSONL telemetry                            |
| R5        | Stream video to a specific IP                               | stream server                              |
| R6        | Store video locally                                         | recording path                             |
| R7        | GUI for monitoring                                          | mission-control SPA                        |
| R8        | Dataset generation for detection, pose, HOI                 | dataset engine                             |
| R9 (opt.) | Orientation-agnostic 3D HMR relative to the rack            | HMR, calibration, rack frame;              |
| D1        | Trained model on an offline standalone system               | All - no cloud API, no internet at runtime |

Every requirement terminates in at least one component and every component traces back to at least one requirement — there is no orphan functionality, a property worth checking, since unjustified components consume the thermal budget the justified ones need.

## 17. Novelty, Limitations, Roadmap and Conclusion

### 17.1 Novelty and consistency audit

Claims are stated at the level they can be defended — no individual algorithm is claimed as invented. The systems-level contributions: (N1) orientation-agnosticism reframed as a coordinate-frame problem — expressing kinematics in a rack-fixed frame  $R$  so thresholds are orientation-independent by construction, not by learned invariance; (N2) camera→rack calibration as a procedural-validation substrate with continuous fiducial monitoring; (N3) metric 3D geometry with stated uncertainty, not classifier confidence, as the evidence type for validation; (N4) an uncertainty-coupled thresholding rule ( $\tau \geq 2 \times \text{RMSE}$ ) enforced at startup; (N5) the Digital Twin as the sole inter-agent interface, making the validator testable with no camera, model or human; (N6) an eight-agent architecture with an announced degradation ladder; (N7) a type/interface/data-flow/telemetry/partition-enforced boundary between generative reasoning and authority, not a prompt-level guardrail; (N8) graph-guided validation composed with (not replacing) the FSM via load-time dominator analysis; (N9) rate-domain separation with time-, not frame-, specified debounce; (N10) structured telemetry with embedded evidence enabling deterministic replay; (N11) full offline integration, demonstrable network-disconnected; (N12) uncommanded object motion as a first-class microgravity-specific event with no terrestrial analogue.

A 20-point consistency audit preceded submission and is recorded in full in the source document; representative resolutions: five rate domains disambiguate previously conflicting FPS claims (§8.3); the debounce specification was corrected from frame-count to time-based after being found to yield 1.0–1.5 s instead of the intended 0.5 s at the real inference rate; an inherited "3,000,000:1 compression ratio" claim was withdrawn as not a compression ratio and replaced with a defensible, measurable framing; the agent count (four vs. eight across earlier drafts) was reconciled to eight, authoritative; and every SIH requirement was re-traced to source components (15.4). No fabricated results, no future functionality presented as implemented, and no unsupported "exact distance" claims remain.

### 17.2 Advantages

Verdicts are explainable (evidence record per transition) and reproducible (deterministic replay resolves disputes). Orientation independence is by construction, not learned. Downlink is bandwidth-proportionate to events, not video duration. No hardware modification to payload items is required. Degradation is graceful and announced — capability shown always matches capability available. Errors are biased toward omission, never accusing an astronaut of a mistake not made. The blackboard decouples components so the safety-critical validator is independently, exhaustively testable. Generative reasoning adds natural-language value with zero authority leakage. Operation is fully offline, including during loss-of-signal. Record and stream paths are independent, so a network fault cannot gap the scientific record. New experiments are a configuration file, not a code change. Temporal logic is thermally invariant. Uncertainty is propagated end to end, so gates automatically distrust degraded perception.

### 17.3 Limitations

Stated first, because it is the most important: no model in this system has a measured accuracy figure from our own training — the architecture is specified and partially implemented, but the measurements that would validate it have not yet been taken.

| Group                 | Limitations                                                                                                                                                                                                                         | Mitigation today                                                                                                                                                                                  |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Perception & geometry | Monocular depth is estimated, not measured, and its error grows with distance; HMR hand-joint error limits HOI precision; a single fixed camera has no second viewpoint for occlusion; calibration drift biases geometry coherently | Contact-anchored depth, relative-distance preference, uncertainty-coupled thresholds; confidence gating and outlier rejection; explicit HOLD rather than guessing; continuous fiducial monitoring |
| Data & validation     | No real microgravity data; synthetic randomisation addresses appearance, not free-fall dynamics; unmeasured accuracy (above)                                                                                                        | Domain randomisation, semi-supervised adaptation, real-only test split; parabolic-flight/analogue data is roadmap Phase 9                                                                         |

| Group             | Limitations                                                                                                                          | Mitigation today                                                                                                                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Compute & latency | Full-rate full-resolution inference is not sustainable; alert latency is bounded by inference + debounce + TTS                       | Rate decimation, ROI gating, proactive throttle avoidance; phrase caching, asymmetric debounce                                       |
| Scope             | Generative advisory output can be wrong; single-subject assumption; procedure-specific models/protocols need new data per experiment | Architecturally advisory-only with consistency checks; nearest-to-rack selection; configuration-driven protocols, reusable HOI layer |
| Flight readiness  | No hardware validation on target edge platform; no parabolic-flight/drop-tower testing; no radiation or space qualification          | Development/target/flight separation stated throughout, never conflated; roadmap Phases 7, 9, 10                                     |

## 17.4 Scalability

The architecture scales in the dimensions that matter because validation is nearly free and perception is the constrained resource: more procedure steps, branching, additional racks, and longer sessions cost negligible runtime ( $O(1)$ -per-frame FSM, load-time dominator analysis, bounded buffers already in place). What does not scale as cleanly: per-experiment dataset effort (new objects/interactions need new labelled data — synthetic generation reduces but does not eliminate this), simultaneous subjects on one edge module (HMR cost is per-subject), and very large in-frame object counts (bounded in practice by rack workspace size). The correct shape is preserved throughout: procedural complexity is unbounded and cheap; perceptual complexity is bounded and expensive.

## 17.5 Future roadmap

| Phase | Content                                                                    | Status          |
|-------|----------------------------------------------------------------------------|-----------------|
| 1     | Webcam+PC prototype: capture, detection, FSM, GUI, TTS, telemetry          | [A] Complete    |
| 2–3   | HMR/HAR/HOI + rack-frame integration; Digital Twin + replay tooling        | [B] In progress |
| 4–6   | Local VLM advisory; IMU+UKF fusion; graph-guided validation                | [A] Complete    |
| 7–8   | Edge (Jetson/TensorRT) deployment; full robustness/fault-injection testing | [C] Planned     |
| 9     | Microgravity analogue testing (parabolic flight, drop tower)               | [E] Future      |
| 10    | Pre-flight / space qualification pathway                                   | [E] Future      |

The critical path runs through §15.2: Phase 2 cannot close until the 2D-to-3D evaluation is executed, since every downstream threshold depends on its result.

## 17.6 SIH demonstration plan

A ~10-minute, ~20-step demo built to make the architecture visible, verdicts verifiable, and honesty evident: network visibly disconnected before anything runs (D1); live detection, HMR and Digital Twin overlays; a correct run through the benchmark; a deliberate out-of-order grasp producing an audible warning; the raw JSONL evidence record shown live; recovery; completion; the LAN stream and local recording shown; an orientation demonstration (inverted/horizontal, verdicts unchanged); and a closing statement naming what is implemented, what is planned, and that §15.2 has not yet been run. The three steps that matter most beyond "it works" are the live evidence record (a verdict decomposed into checkable physical measurements), an occlusion demonstration where the system enters HOLD rather than guessing, and the closing statement itself — a team that states its own limitations first is more persuasive when it then states its achievements.

## 17.7 Conclusion

SIH Problem Statement 26174 asks for an on-board assistant that validates a scientific procedure without ground supervision — which is not principally a perception problem but a trust problem. This system pairs neural perception (detection, orientation-agnostic 3D HMR, metric rack-frame geometry, temporal activity understanding) for capability with a deterministic FSM (finite, enumerable, debounced, replayable, evidence-logged) for authority, mediated by a Digital Twin, with generative reasoning architecturally confined to the advisory side of that boundary. What exists today is a working, fully offline prototype on the benchmark experiment; what does not yet exist — the HMR path, inertial fusion, the local VLM, graph validation, the edge port, and any measured

accuracy figure — is stated as plainly as what does. The single highest-priority outstanding task is §15.2's evaluation, since every operational threshold depends on it. The goal throughout has not been to sound impressive, but to make every decision traceable to a requirement, justified by an argument, bounded by a stated limitation, and checkable by a defined experiment.

## 17.8 References

Full citations appear in the source documentation across seven groups: the SIH problem statement and BAS mission-context material; the team's own prior thesis and system-design documents (primary references for this work); adopted literature methods — PhysAstro-Pose (semi-supervised microgravity pose estimation, §11), Wen et al. on terrestrial-model failure in microgravity (§3.1), AdaSpot (dual-resolution saliency gating, §21), and  $\mu$ Bi-ConvLSTM (candidate HAR architecture, §19); Neutral Body Posture biomechanics from Mir/ISS long-duration flights (§3.2); hardware datasheets (Jetson Orin NX, PIC64-HPSC); third-party flight precedent (POEM-4 payloads on PSLV-C60/SpaDeX, cited as precedent only — conferring no heritage on this system); and standard methods (Zhang camera calibration, Kabsch/Umeyama rigid transform, Julier–Uhlmann UKF, dominator analysis on directed graphs).